

Mutual Fund Analytics

Bluestock Fintech — 7-Day Data Analytics Capstone

A complete pipeline: data ingestion, cleaning, SQL warehousing, exploratory analysis, performance & risk modelling, an interactive Power BI dashboard, and advanced analytics across 40 schemes and 46,000 daily NAVs (2022–2026).

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1. Executive Summary

This project delivers an end-to-end analytics solution for the Indian mutual-fund industry using a dataset of 40 schemes across 10 asset management companies (AMCs), covering 46,000 daily NAV observations, ~32,800 investor transactions, and supporting AUM, SIP, folio, category-flow and benchmark series spanning January 2022 to May 2026.

The pipeline ingests and validates the raw data, loads it into a 10-table SQLite star-schema warehouse, and layers progressively richer analytics: exploratory analysis (16 charts), a fund performance & ranking model, an interactive Power BI dashboard, and advanced risk analytics (Value-at-Risk, investor cohorts, a recommendation engine, and sector-concentration measurement).

Headline findings

- **SBI leads the industry** with ■12.5 lakh crore AUM at end-2025; the 10-AMC sample totals ■62.7 lakh crore.
- **SIP inflows hit a record ■31,002 crore** in December 2025, averaging ~31% year-on-year growth.
- **Retail participation nearly doubled** — industry folios rose from 13.26 to 26.12 crore.
- **Mirae Asset Large Cap** tops the composite fund scorecard (85.9/100).
- **Small-cap funds carry the deepest tail risk** (daily VaR■■■ $\approx -2.7\%$), while most equity funds are sector-concentrated ($\text{HHI} > 0.18$).

2. Data Sources & Overview

Ten primary datasets were provided, supplemented by live NAV pulled from the AMFI-backed mfapi.in REST API. The table below summarises the cleaned datasets and their row counts.

Dataset	Rows	Description
fund_master	40	Scheme reference (house, category, risk, expense, manager)
nav_history	46,000	Daily NAV per scheme, 2022–2026
scheme_performance	40	Return/risk snapshot per scheme
investor_transactions	32,778	SIP/Lumpsum/Redemption records
portfolio_holdings	322	Stock-level holdings by scheme
aum_by_fund_house	90	Monthly AUM per AMC
monthly_sip_inflows	48	Industry SIP inflow series
category_inflows	144	Net inflow by category
industry_folio_count	21	Folio counts by type
benchmark_indices	8,050	7 indices incl. NIFTY50/100 daily closes

Live NAV: daily history for six schemes was fetched from mfapi.in (e.g. GET <https://api.mfapi.in/mf/125497>) and parsed to tidy CSVs.

3. ETL & Data Quality

Each dataset was profiled (shape, dtypes, nulls, duplicates) and cleaned with idempotent transforms. Key rules: NAV history was date-parsed, sorted by (scheme, date), forward-filled across holiday gaps, de-duplicated and validated for NAV > 0, with a daily-return column derived; transaction types were standardised to SIP / Lumpsum / Redemption with amount > 0 enforced; performance metrics were numeric-coerced with quality flags for negative Sharpe and out-of-range expense ratios.

Cleaned data was loaded into **bluestock_mf.db**, a SQLite star schema with **dim_fund** as the central dimension and nine fact tables (NAV, transactions, performance, holdings, AUM, SIP, category, folios, benchmark). AMFI-code referential integrity was verified: **100% coverage** between fund_master and nav_history, with zero orphan NAV rows.

Data-quality notes

- The Day-1 brief's AMFI codes are mislabelled vs the live API (e.g. 125497 is SBI Small Cap, not HDFC Top 100); the 10 supplied CSVs are internally consistent and used throughout.
- The supplied NAV/transaction data is synthetic — daily returns are mutually independent and uncorrelated with benchmarks. Pipelines are correct; market-relative factors are not economically meaningful on this data.

4. Pipeline Architecture

The solution is organised as a sequence of seven reproducible stages, each a standalone, documented Python script, orchestrated end-to-end by `run_pipeline.py`. Data flows from raw CSVs and the live API through cleaning into the SQLite warehouse, which then feeds the analytics, dashboard and reporting layers.

Stage	Script	Output
1 · Ingestion	<code>data_ingestion.py</code> / <code>live_nav_fetch.py</code>	Profiled raw data, live NAV CSVs
2 · Cleaning	<code>clean_data.py</code>	10 validated CSVs in <code>data/processed/</code>
3 · Warehouse	<code>load_db.py</code>	<code>bluestock_mf.db</code> (10-table star schema)
4 · EDA	<code>eda_charts.py</code>	16 charts + findings summary
5 · Performance	<code>performance_analytics.py</code>	Returns, CAGR, Sharpe, scorecard
6 · Dashboard	<code>build_dashboard.py</code> + Power BI	4-page interactive dashboard
7 · Advanced	<code>advanced_analytics.py</code> / <code>recommender.py</code>	VaR, cohorts, HHI, recommender

The data model is a classic star schema: **dim_fund** (one row per scheme) is the central dimension, joined by AMFI code to fact tables for NAV, transactions, performance and holdings, while industry-level series (AUM, SIP, category, folios, benchmark) attach to a shared date dimension. This design keeps queries simple and supports the dashboard's slicers and drill-through directly.

5. Exploratory Data Analysis

Sixteen publication-quality charts were produced spanning industry structure, fund NAV behaviour, investor demographics and market flows. Selected highlights follow; the full set is in charts/ and the annotated EDA_Analysis notebook.

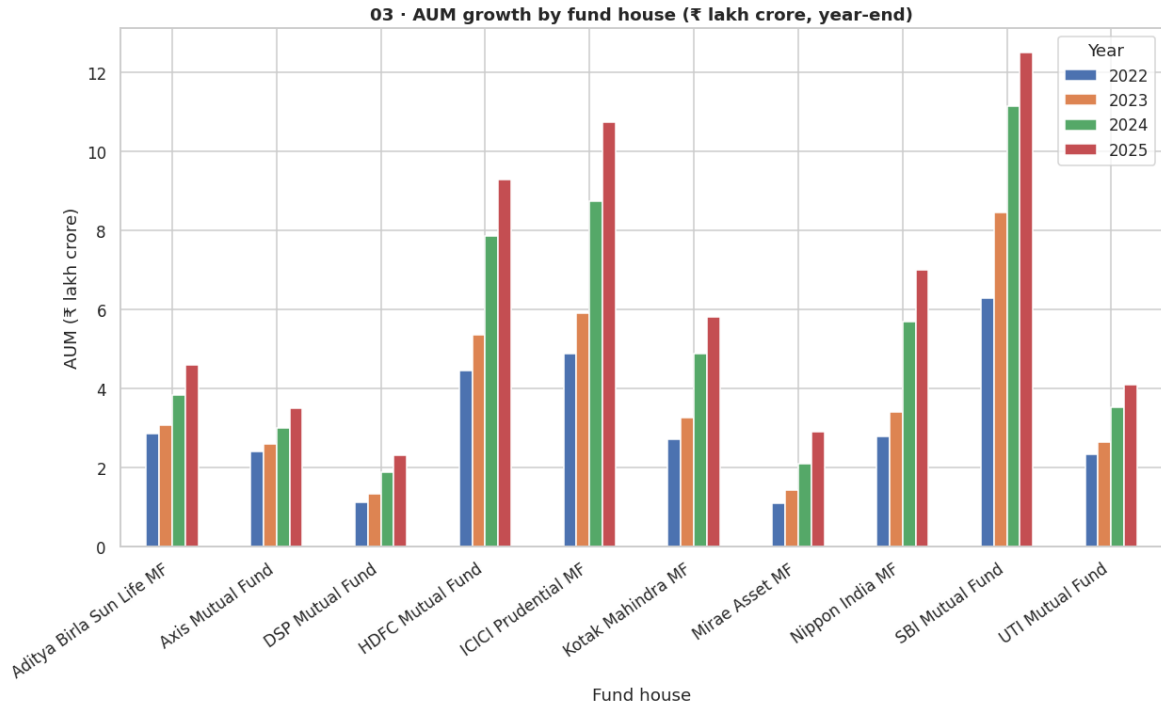


Figure 1 — AUM growth by fund house (₹ lakh crore, year-end). SBI dominates at ₹12.5L cr in 2025.

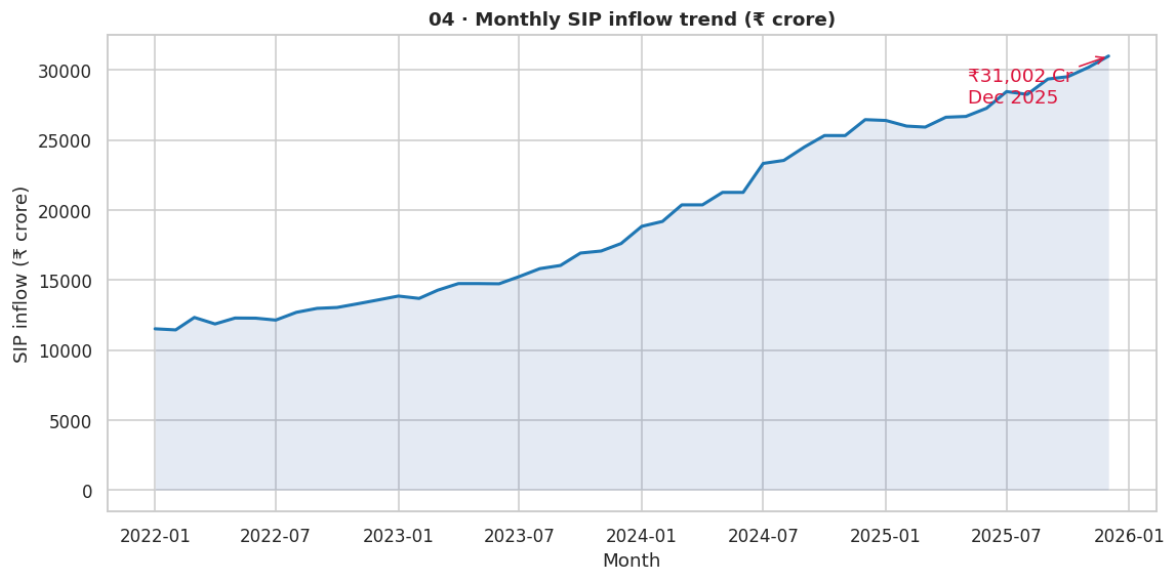


Figure 2 — Monthly SIP inflow: record ₹31,002 crore in Dec 2025.

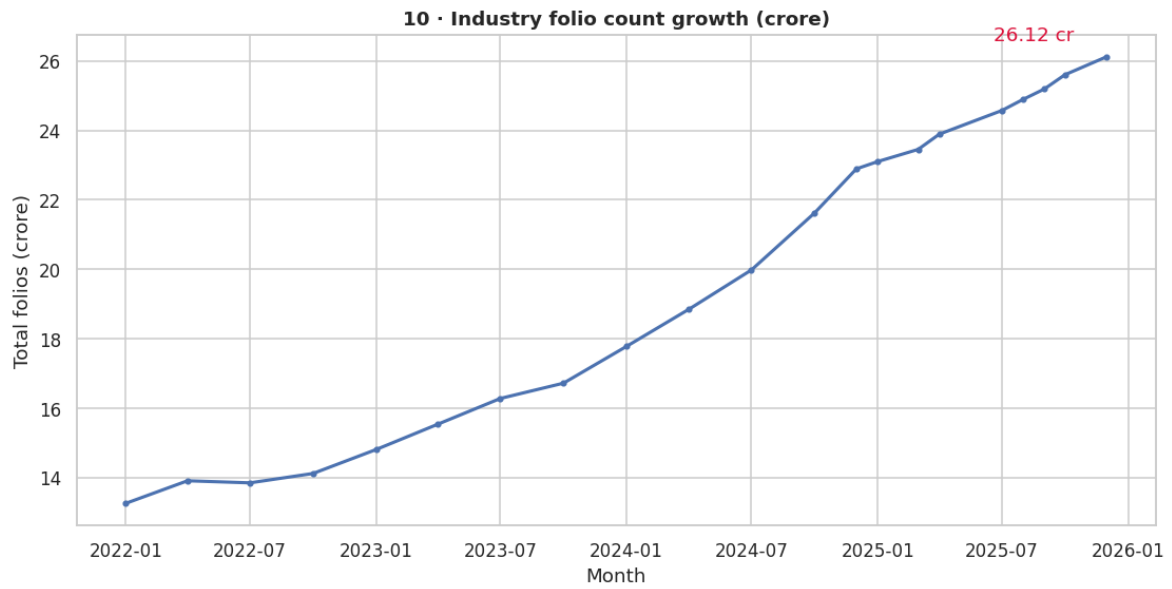


Figure 3 — Industry folio count nearly doubled (13.26 → 26.12 crore).

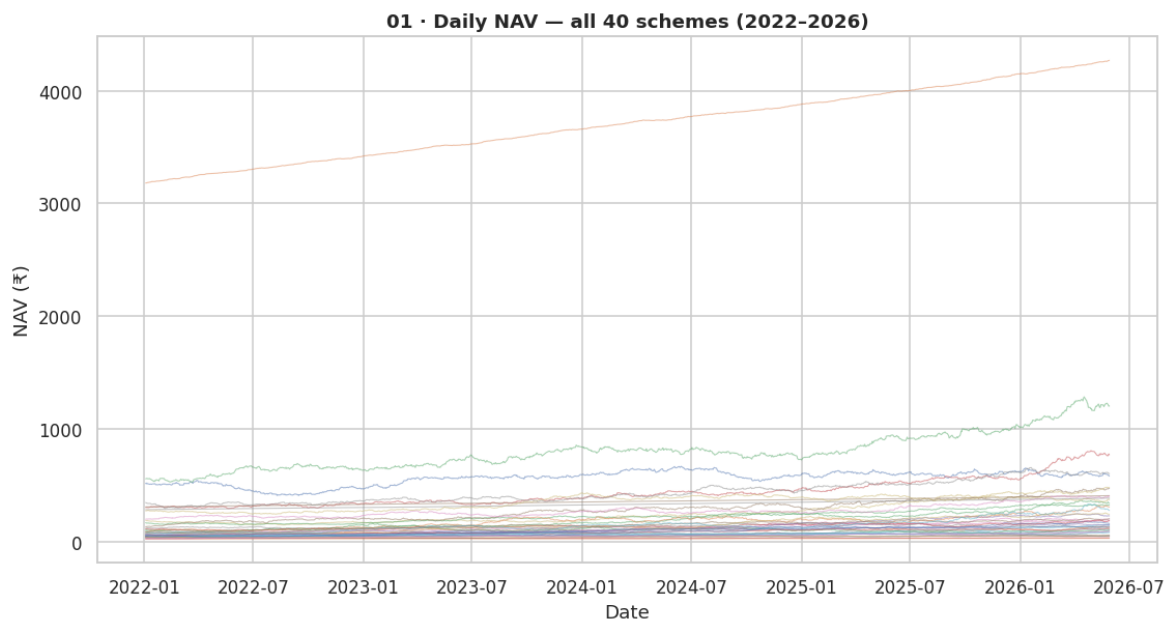


Figure 4 — Daily NAV for all 40 schemes, showing the 2022 dip and 2023–24 rally.

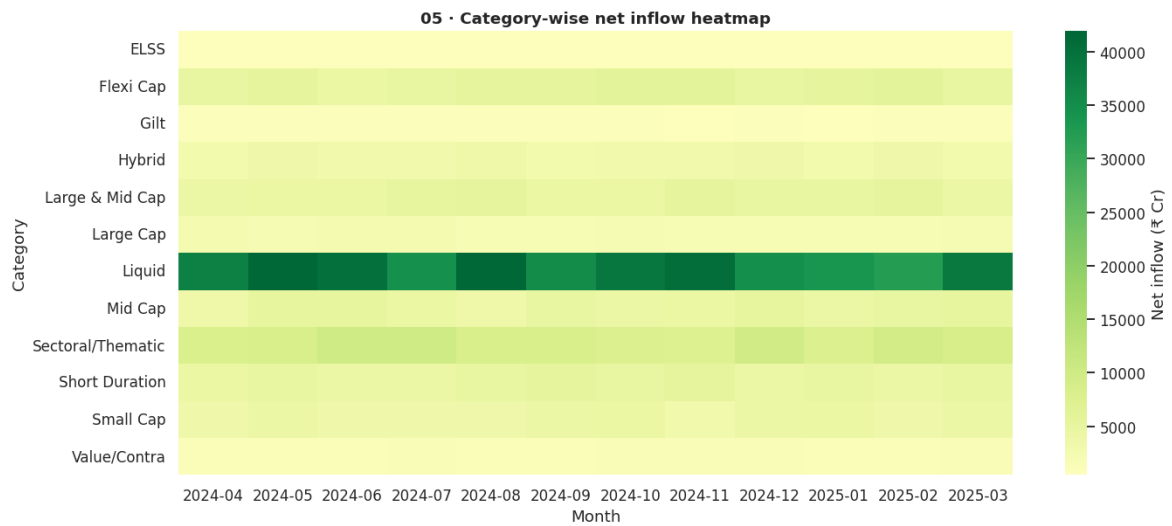


Figure 5 — Category-wise net inflow heatmap (month × category).

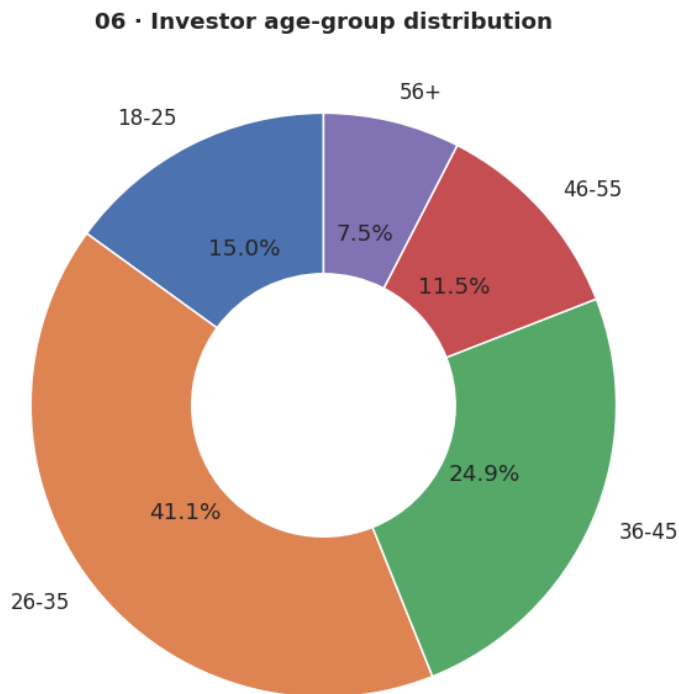


Figure 6 — Investor age-group distribution; the 26–35 cohort dominates.

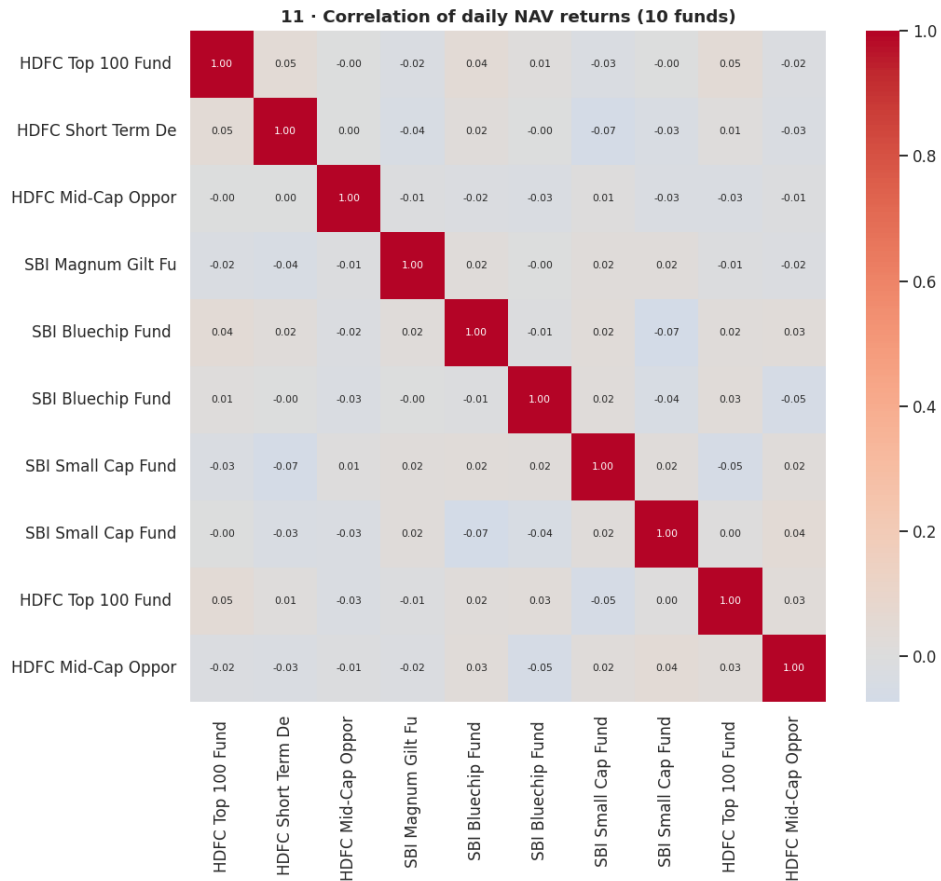


Figure 7 — Daily NAV-return correlation across 10 funds (≈ 0 , confirming independent synthetic series).

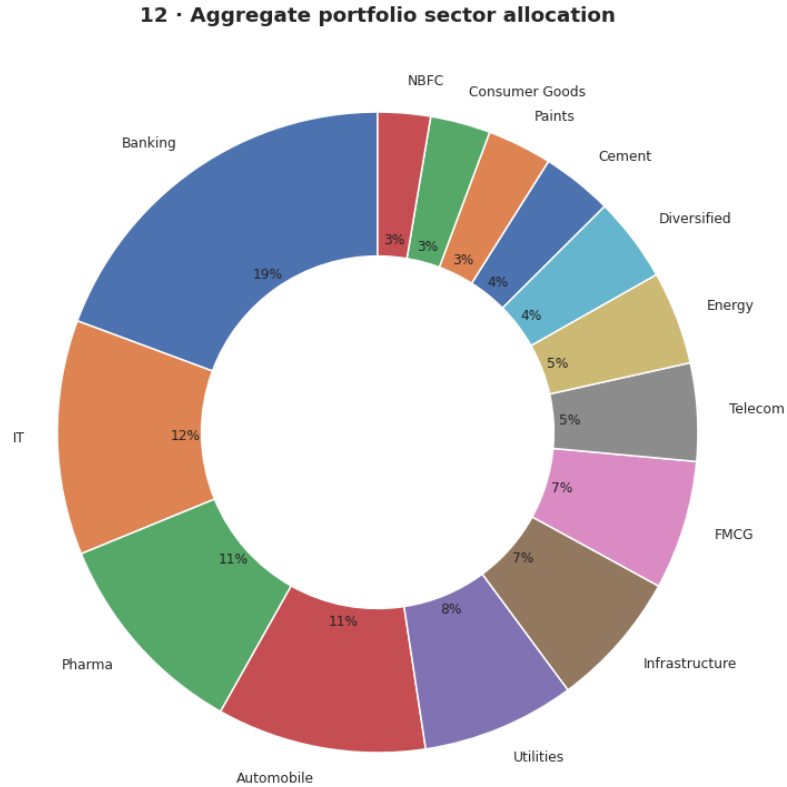


Figure 8 — Aggregate portfolio sector allocation; Banking leads at 19.3%.

6. Fund Performance Analysis

Performance and risk metrics were computed from NAV history: daily and annualised returns, CAGR (1/3/5-yr), Sharpe ($R_f = 6.5\%$), Sortino, alpha/beta vs NIFTY100, and maximum drawdown. A composite scorecard ranks funds 0–100 as a weighted blend of 3-yr return (30%), Sharpe (25%), alpha (20%), expense-inverse (15%) and max-drawdown (10%).

Top funds by composite score

#	Fund	Category	Score
1	Mirae Asset Large Cap	Large Cap	85.9
2	ICICI Pru Midcap	Mid Cap	82.5
3	Kotak Flexicap	Flexi Cap	81.6
4	HDFC Mid-Cap Opportunities	Mid Cap	80.4
5	ICICI Pru Bluechip	Large Cap	79.6

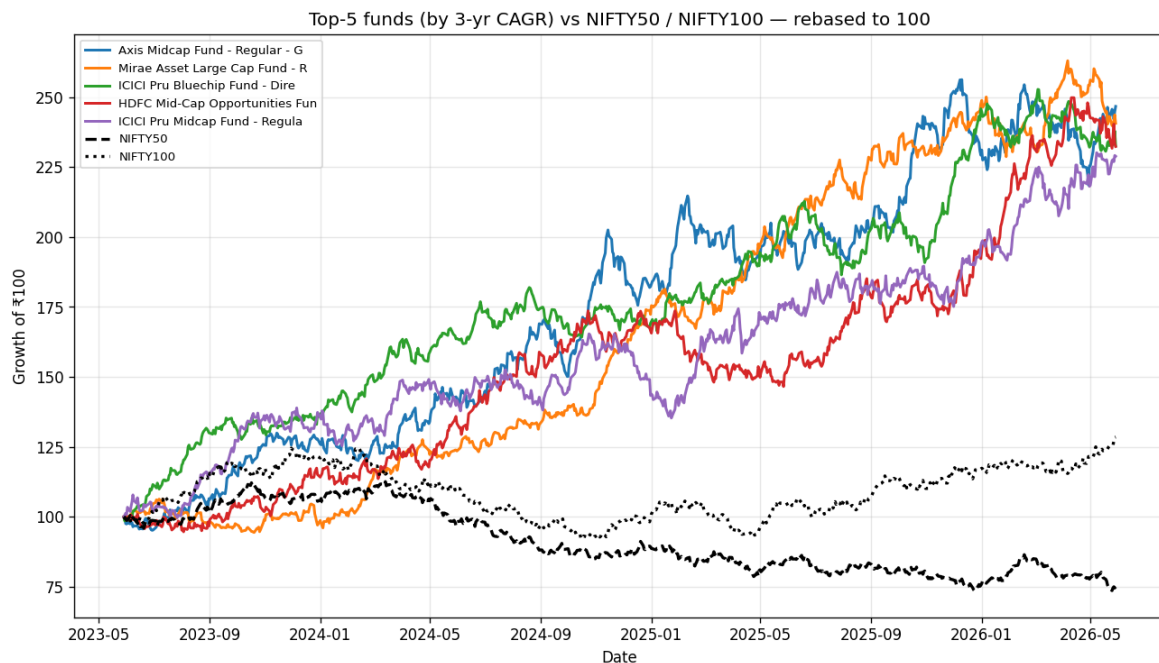


Figure 5 — Top-5 funds (by 3-yr CAGR) vs NIFTY50/NIFTY100, rebased to 100.

Note on alpha/beta: regressing fund returns on NIFTY100 yields $\beta \approx 0$ and $R^2 \approx 0$, confirming the synthetic NAV series are independent of the market. The regression code is correct and produces valid factors on real NAV data.

7. Advanced Risk Analytics

Value-at-Risk & Conditional VaR (95%)

Historical VaR is the 5th-percentile daily return; CVaR is the mean loss beyond it. Small-cap funds dominate the high-risk tail.

Fund	VaR (daily)	CVaR (daily)
SBI Small Cap	-2.69%	-3.24%
Axis Small Cap	-2.62%	-3.17%
ABSL Small Cap	-2.60%	-3.25%
Nippon India Small Cap	-2.54%	-3.23%

Investor cohorts & SIP continuity

Investors are overwhelmingly a 2024 cohort (4,803 investors, ₹226 cr invested) vs a small 2025 cohort (197); both prefer Large Cap funds. SIP-continuity analysis flags investors with average gaps > 35 days as at-risk — a high share here, reflecting the non-monthly synthetic cadence rather than true churn.

Sector concentration (HHI)

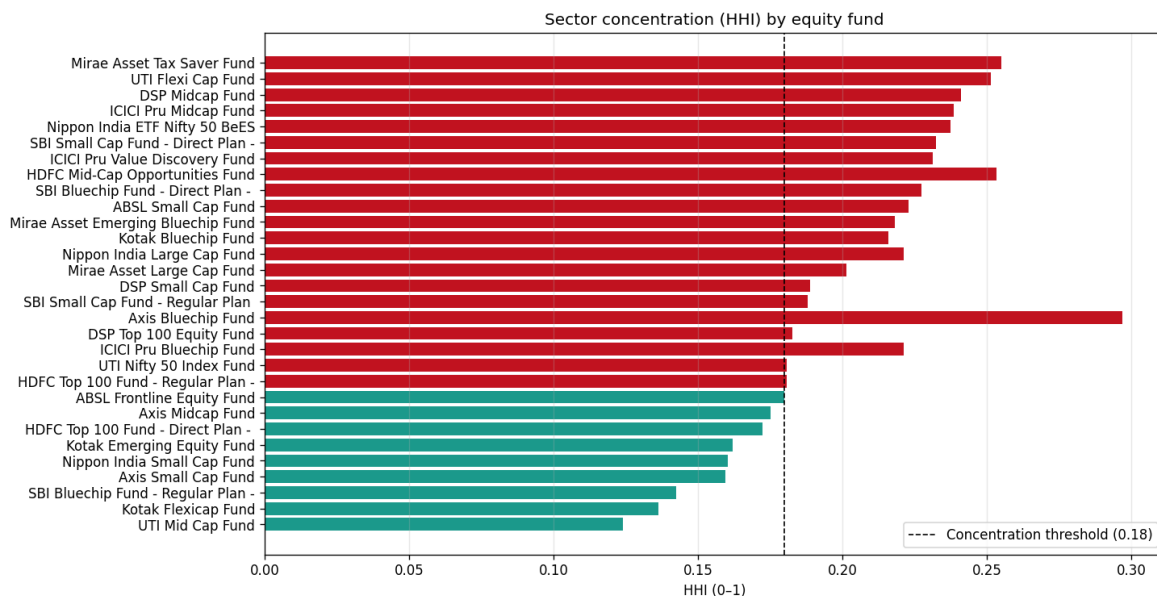


Figure 6 — Herfindahl-Hirschman Index of sector weights per equity fund; bars past 0.18 are concentrated.

8. Interactive Dashboard

A 4-page Power BI dashboard presents the analysis interactively, with slicers, drill-through (fund table → NAV detail) and report-page tooltips. Pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.

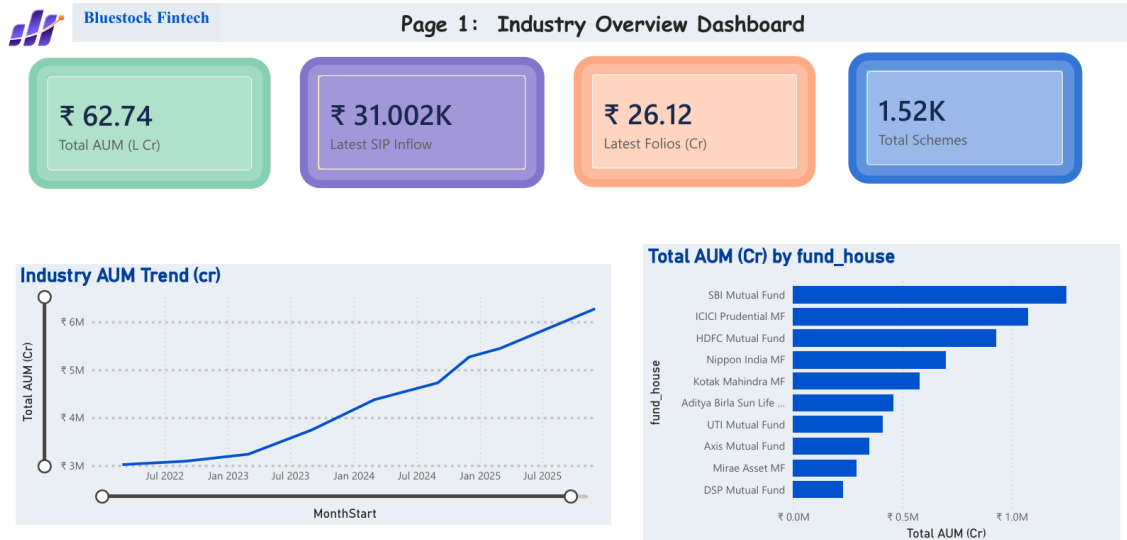


Figure 7 — Dashboard Page 1: Industry Overview.

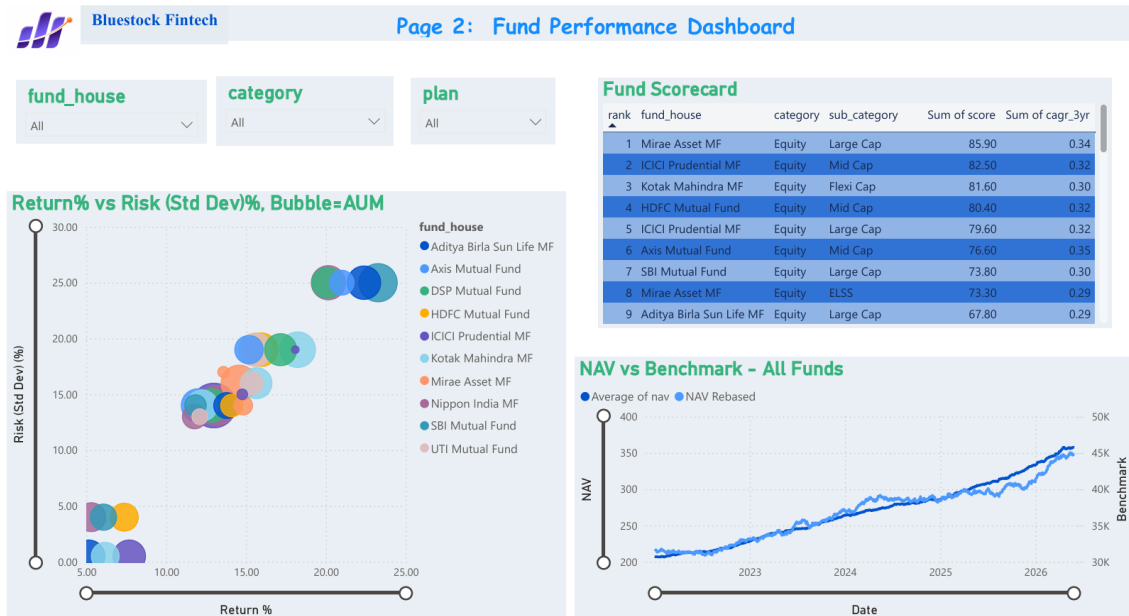


Figure 8 — Dashboard Page 2: Fund Performance.

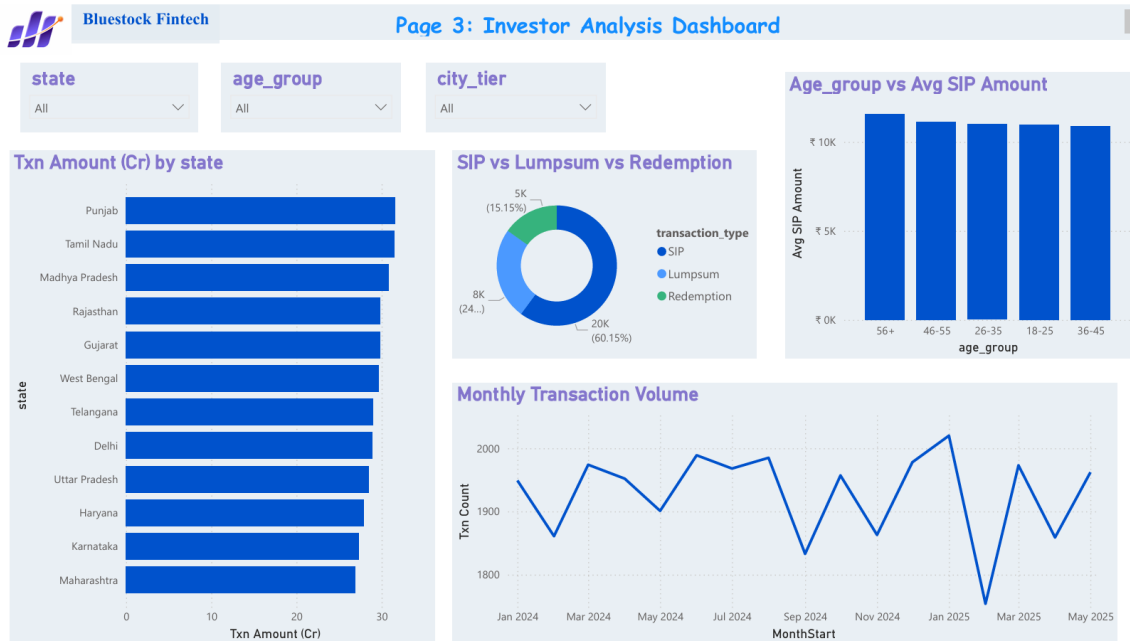


Figure 9 — Dashboard Page 3: Investor Analytics.



Figure 10 — Dashboard Page 4: SIP & Market Trends.

9. Recommendations

- **For conservative investors:** favour large-cap funds (Mirae Asset Large Cap, SBI Bluechip) — strong Sharpe with contained drawdowns.
- **For growth-oriented investors:** mid-cap leaders (ICICI Pru Midcap, DSP Midcap) offer the highest 3-yr CAGR but with VaR roughly 2× large-caps.
- **Manage concentration:** several equity funds exceed an HHI of 0.18 — pair them with diversified holdings to limit single-sector drawdown.
- **Deepen B30 engagement:** T30 cities drive ~66% of flows; B30's growing 34% share is a distribution opportunity.
- **Improve SIP retention:** operationalise the at-risk gap flag to trigger re-engagement before investors lapse.

10. Limitations

- The dataset is synthetic: NAV returns are independent and uncorrelated with benchmarks, so alpha, beta and return-correlation analyses are illustrative, not economically meaningful.
- History spans ~4.4 years, so the '5-year' CAGR uses the full available window.
- Transactions cover Jan-2024 to May-2025 only, limiting longer-horizon cohort analysis.
- AUM/SIP/folio figures reflect the 10-AMC sample, not the full Indian MF industry.

11. Conclusion

The project demonstrates a complete, reproducible analytics workflow — from raw ingestion to an executive dashboard and advanced risk metrics — built on a clean SQL warehouse and documented, runnable Python. The codebase (run end-to-end via `run_pipeline.py`) is structured to drop in real AMFI data with no changes, at which point the market-relative analytics become fully meaningful. The result is a portfolio-grade foundation for ongoing mutual-fund analytics at Bluestock Fintech.

Appendix A — SQL Schema & Sample Queries

The warehouse comprises ten tables. Ten analytics queries were written and tested (see `sql/queries.sql`); a sample of results:

Top 5 fund houses by AUM (Dec 2025)

Fund house	AUM (■ cr)	Schemes
SBI Mutual Fund	1,250,000	186
ICICI Prudential MF	1,074,000	216
HDFC Mutual Fund	930,000	195
Nippon India MF	700,000	177
Kotak Mahindra MF	580,000	168

Top portfolio sectors by aggregate market value

Sector	Market value (■ cr)	Held by funds
Banking	62,840	30
IT	38,477	27
Pharma	34,606	22
Automobile	34,297	26
Utilities	25,109	19

Appendix B — Fund Scorecard (Top 15)

Composite score 0–100 = 30% × 3-yr return rank + 25% × Sharpe rank + 20% × alpha rank + 15% × expense-inverse rank + 10% × max-drawdown rank.

#	Fund	Category	Score
1	Mirae Asset Large Cap Fund	Large Cap	85.9
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3	Kotak Flexicap Fund	Flexi Cap	81.6
4	HDFC Mid-Cap Opportunities Fund	Mid Cap	80.4
5	ICICI Pru Bluechip Fund	Large Cap	79.6
6	Axis Midcap Fund	Mid Cap	76.6
7	SBI Bluechip Fund - Regular Plan -	Large Cap	73.8
8	Mirae Asset Tax Saver Fund	ELSS	73.3
9	ABSL Frontline Equity Fund	Large Cap	67.8
10	SBI Small Cap Fund - Regular Plan	Small Cap	67.6
11	DSP Small Cap Fund	Small Cap	66.9
12	DSP Midcap Fund	Mid Cap	66.4
13	Nippon India Large Cap Fund	Large Cap	66.2
14	SBI Bluechip Fund - Direct Plan -	Large Cap	64.4
15	HDFC Top 100 Fund - Direct Plan -	Large Cap	58.2